1 Planning (15 marks)

Defining the problem (3 marks)

P (cos) θ is the independent variable, or vary (cos) θ. [1]

P *P* is the dependent variable, or measure *P*. [1]

P Keep the speed of the air constant.
Allow keep power to the fan/hairdryer constant. [1]

Methods of data collection (5 marks)

M Labelled diagram showing method to produce air flow in line with turbine. Method of producing "wind" must be labelled. [1]

M Circuit connecting turbine to lamp with ammeter and voltmeter connected correctly.
No additional power supplies in the lamp circuit. [1]

M $P = IV$. Do not allow I^2R or V^2/R unless it is clear that R is determined from V/I.
Allow wattmeter or joule meter and stopwatch. [1]

M Measure angle with protractor or use rule to measure appropriate distances. [1]

M Ensure that there are no other draughts or airflows. [1]

Method of analysis (2 marks)

A Plot a graph of *P* against cos θ. [1]

A *k* = gradient. [1]

Safety considerations (1 mark)

S Precaution linked to avoiding air flow entering eyes or avoid moving blades. [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Use of large wind speed to gain measurable readings.
2 Use of low wattage/low resistance lamp or turbine with low friction.
3 Additional detail on measuring (cos) θ – correct angle must be determined.
4 Wait until airflow/turbine/meter readings constant.
5 Avoid turbulence or reflection of air flow.
6 Ensure distance from fan to turbine is constant.
7 Relationship is valid if the graph is a straight line passing through the origin.
8 Method to check that wind speed is constant.

Do not allow vague computer methods.

[Total: 15]

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